Construction Robot Digital Twin

Autonomous Navigation to a Wall-Mounted AprilTag

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CIVIL 8025 Project 4

ROS2 + Gazebo + Nav2 + Local Web Dashboard

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Four-wheel inspection robot, physics-based simulation, SLAM-ready map workflow, and browser-based operator interface.

Presentation Roadmap

What I built

- A ROS2 construction robot digital twin
- Gazebo world with collision constraints
- Nav2 mission from start point A to AprilTag target

What I will show

- Robot model and control interfaces
- Navigation, SLAM, telemetry, and safety
- Final video demo after the slides

Project Goal

Background

Construction sites are dynamic, cluttered, and safety-critical. Autonomous mobile robots can support inspection work by navigating through site-like areas, checking target locations, and giving operators live feedback without sending people into every space manually.

Inspection task

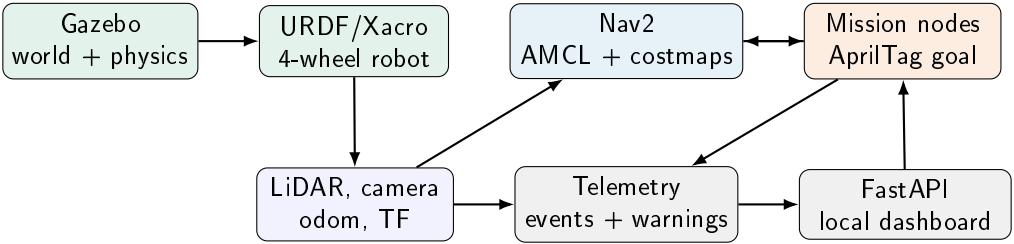
The robot starts at point **A**, autonomously navigates through a construction-like indoor environment, approaches a wall-mounted **AprilTag**, and reports the mission as complete.

- Mobile robot must obey physical and collision constraints.
- Operator can switch between automatic mission and manual control.
- Dashboard gives live feedback instead of

Task pipeline

1. Start from A.
2. Plan around obstacles with Nav2.
3. Approach the wall AprilTag.

Digital Twin Architecture



The dashboard is an experience layer over ROS topics, actions, and services; it does not replace Gazebo or RViz engineering tools.

Robot and Environment Model

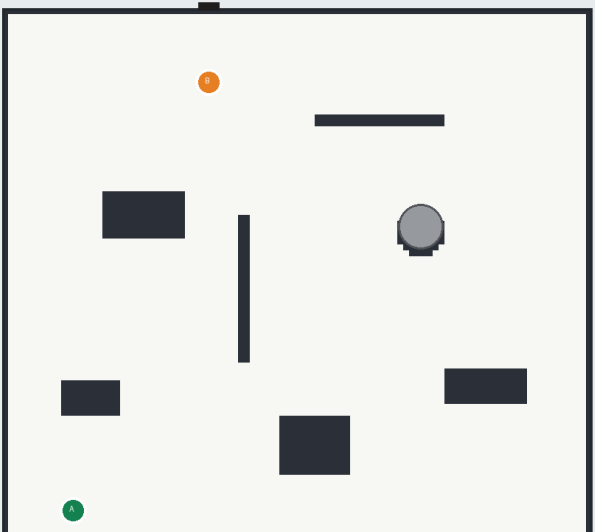
Robot

- Four-wheel differential/skid-steer base
- Rear left/right wheels driven by one Gazebo diff-drive plugin
- Passive front wheels for stable contact
- Pan/tilt camera gimbal with joint limits
- LiDAR, RGB camera, odometry, and TF

Construction scene

- Walls and floor with collision bodies
- Barriers, material stacks, tool chest, rebar cage
- Simple cylinder column for stable obstacle geometry
- Wall-mounted AprilTag target
- Static map plus SLAM workflow

Autonomous Mission



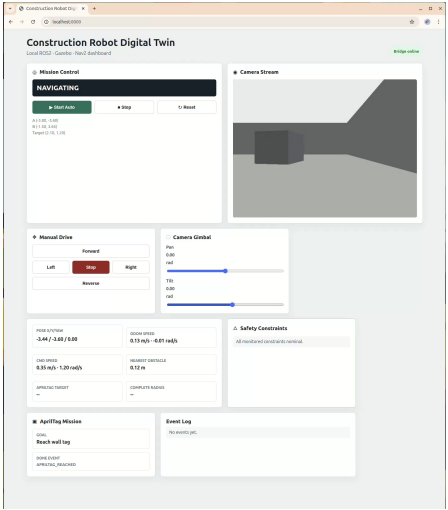
Mission sequence

1. Publish AMCL initial pose at A
2. Wait for `NAV2_READY`
3. Send B as `NavigateToPose`
4. Approach the AprilTag wall
5. Publish `APRILTAG_REACHED`

Map information

- Static map: `room_map.yaml`
- Resolution: 0.1 $m/cell$
- Size: 100×100 cells

Web Dashboard



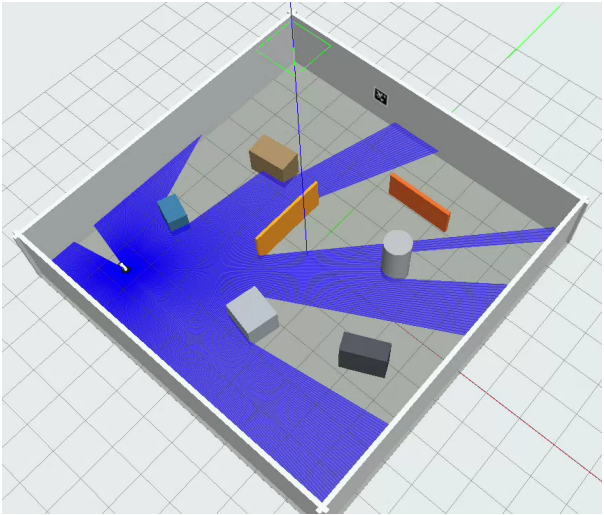
Operator controls

- Start Auto, Stop, Reset
- Manual forward, reverse, left, right
- Camera pan and tilt sliders
- Local camera stream endpoint

Live feedback

- Mission state and event log
- Pose, velocity, command velocity
- Nearest LiDAR obstacle range
- AprilTag target information
- Safety warning panel

SLAM and Map Workflow



Mapping mode

- Launch Gazebo with SLAM Toolbox
- Drive manually from the dashboard
- RViz shows the LiDAR-built `/map`
- Save the map for later Nav2 runs

Navigation mode

- Load `room_map.yaml`
- AMCL estimates `map -> odom`
- Diff-drive plugin publishes `odom -> base_link`
- Nav2 plans through global/local costmaps

Safety and Physical Constraints

Model-level constraints

- URDF collision geometry and inertia
- Wheel contact and friction parameters
- Gimbal pan/tilt joint limits
- Gazebo collision for walls and obstacles

Runtime constraints

- Nav2 obstacle and inflation layers
- Maximum linear and angular speeds
- Nearest obstacle distance warning
- Emergency `STOP` publishes zero velocity

Safety evidence is visible in code, costmaps, telemetry, and the dashboard.

Implementation Highlights

Simulation

`my_robot_sim`
Gazebo world, wall AprilTag, obstacles, robot spawn.

Navigation

`my_robot_navigation`
Nav2 parameters, map, waypoints, RViz, SLAM launch.

Mission + Web

`my_robot_mission`
Readiness checks, events, telemetry, manual control, markers.

- `START_AUTO` waits for Nav2 action server and `map -> base_link`.
- The mission reports failures explicitly using `NAV2_NOT_READY` events.
- The system can be run reproducibly with Docker and tmux.

Limitations and Future Work

Current limitations

- AprilTag is currently a visual target and known map goal.
- Dashboard shows telemetry, but RViz remains stronger for detailed costmap inspection.
- Real-world deployment would need calibration and sensor noise handling.

Next steps

- Add real AprilTag image detection overlay.
- Stream LiDAR/costmap view directly into the dashboard.
- Use tag pose estimation for final visual servoing.

Conclusion

What the project demonstrates

- A complete construction robot digital twin with physics, sensors, navigation, and operator interface.
- Autonomous navigation from A to a wall-mounted AprilTag target.
- Evidence of collision modelling, costmap-based planning, telemetry, and safety constraints.
- A reproducible Docker/tmux workflow for demonstration and future development.

Next: recorded mission video.